

Sharing scientific data for human-AI collaborations

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Abstract

Scientific data analysis is increasingly dependent on agentic AI, but most best practices for exchanging data are based on the strengths and weaknesses of humans. How can data be packaged in a future-proof way that is optimised for both humans and their machine collaborators? Here, we propose the "data capsule" as a basic unit of scientific data sharing, a single file which stores the data and metadata for an experiment together with the code used to collect, preprocess, and load them. We demonstrate an AI-ready experiment management system for creating and organising data capsules in systems neuroscience. Our system has undergone over 14 person-years of use as well as extensive use by AI scientists. We show how the computational context provided in data capsules helps AI agents interpret data. Overall, data capsules help humans and AI agents work together to accurately and efficiently analyse scientific data.

Note: Information was redacted from this preprint. The data, methods, and software described here do not currently appear on the internet, but the AI agents in our study extensively searched the internet trying to find the source of the data they were provided. Therefore, details were redacted from this preprint so that follow-up experiments during the review process will not be compromised.

The unredacted version of this preprint, as well as the full source code of our software and experimental data, will be posted in full after journal acceptance. They are also available from the author on request.

Introduction

Scientific data should be packaged in a way that facilitates accurate and efficient use, but the strategies for sharing data with humans may differ from those for sharing data with AI agents. Humans work best when data are in a standardised format [1–3], with detailed textual descriptions of processing pipelines [4–7]. When AI agents analyse data, they are not limited in the same way as humans; they can easily access data in any format [8–10] and quickly trace an analysis pipeline through complex and disjointed code [11]. However, questions remain about the correctness of analysis performed by AI [8, 12–15]. Agents often give incorrect results if they do not have access to the meaning or interpretation of data [8, 16, 17], details which may be trivial to humans or easy to clarify from the scientist who collected the data. Data should be shared in a way that addresses the limitations of both humans and AI agents.

The practice of hand-curating data before sharing is not sustainable. Researchers often curate or “clean up” their data before sharing to make it easier for others to use, which may involve reformatting data, renaming variables, rewriting code, and trying to describe their analysis pipeline in prose to the best of their memory. This takes an inordinate (sometimes prohibitive) amount of time, and has the potential to introduce errors or omissions at each stage [18–21]. Delegating curation tasks to AI agents only makes the problem worse by crystallising the agent’s misunderstandings and mistakes directly into the shared data. For both human and AI agents, such mistakes may be detrimental to scientific results. Ideally, data should be packaged together with the information needed to understand them.

Here, we propose the “data capsule” as a basic unit of scientific data sharing. Data capsules store data and metadata together with the code used to collect, preprocess, and interpret them. Capsules can be generated automatically before data analysis, so that the same capsule may be used for both primary data analysis and widespread distribution. We present a software suite, [software_name], for generating and managing data capsules in systems neuroscience. [software_name] is AI-ready, allowing many of the most difficult aspects of preparing capsules to be performed automatically or in natural language. We show how the context provided by a [software_name] data capsule facilitates agentic AI analysis, and how the lack of such context can introduce errors or uncertainties in how an AI agent interprets data. Data capsules require no extra data curation work at the time of publication, since existing capsules can be directly uploaded to a public data repository. Therefore, data capsules provide a practical way to preserve the meaning of scientific data when shared with humans and AI agents.

Results

Data capsules

Data capsules store data, metadata, and code together. A capsule consists of a single file that contains the preprocessed data together with the raw source code of the pipelines used to gather and preprocess them (Figure 1). It also contains the source code needed to load the data directly from the capsule, the code to assemble the capsule itself, and any relevant metadata (Supplement 1). A capsule serves as a record of provenance and usage, two important elements for accurate and efficient downstream data analysis [7, 18].

Three critical pieces of code should be included in the capsule. First, the capsule should

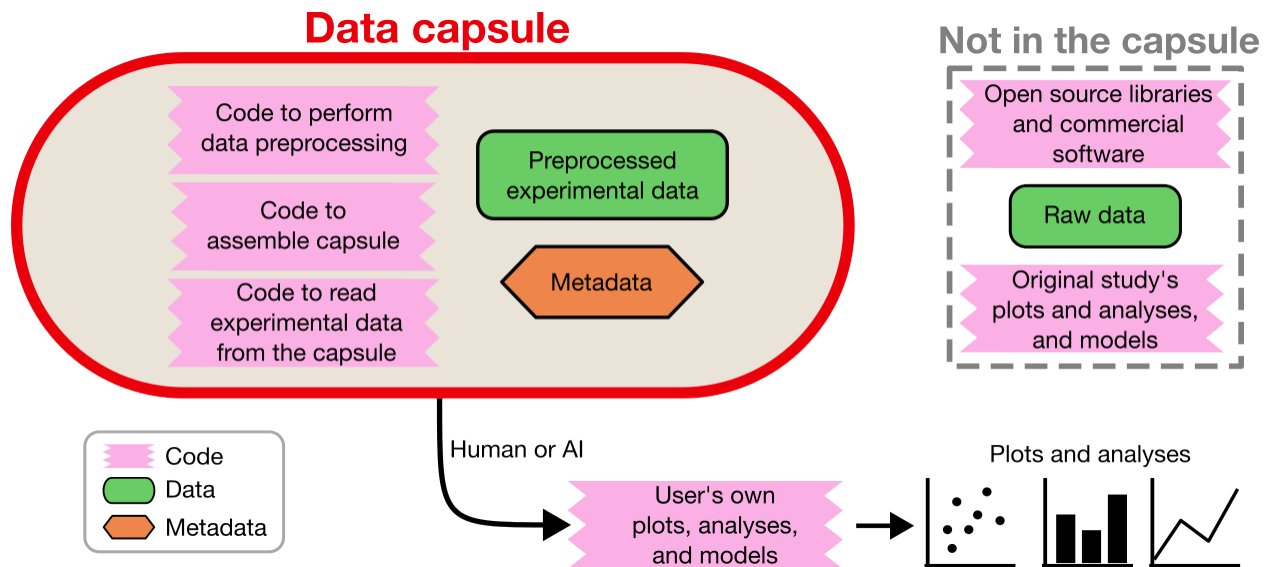


Figure 1: Structure of a data capsule. A capsule stores preprocessed experimental data, meta-data, and code for preprocessing, capsule assembly, and data access. Raw data, external software, and the original study's plots, analyses, and models are not included in the capsule. The capsule provides the context needed for humans or AI agents to perform their own analyses.

include all preprocessing code which was used to convert the data from the raw form to the form stored in the capsule. This code is intended to be used as a reference to show provenance rather than run directly. Second, the capsule should include the code that was used to assemble the capsule itself; this also shows provenance. Third, the capsule should include code to load the data directly from the capsule into a convenient form. This code should consist of functions that can be run directly by the user, and thus, this code may need to be extracted from the capsule first if a local copy is not available. Capsules don't need to include code associated specifically with a publication, such as analyses, models, or plotting. They also don't need to include full copies of commercial or open source software, as is common in reproducible research approaches [22, 23]; the version number and configuration are sufficient for a human or agent to locate this software on the internet. These three pieces of code demonstrate the data's provenance and usage.

To ensure the code and data remain consistent, capsule construction should ideally be automated. This allows a new capsule to be easily generated when new data are available, and regenerated if the preprocessing pipeline is modified. If capsules are constructed or updated by hand, code and data can become out of sync [18, 22, 24–28]. Automating data curation in this way is widely recognised as a way to reduce bugs and ensure data provenance [18, 22]. Being able to easily regenerate capsules also makes it easier to use the same capsule for both primary data analysis and distribution.

Data capsules for systems neuroscience

We created a software suite to easily create, manage, and analyse data capsules for systems neuroscience. This software, which we call [software_name], generates data capsules from *in vivo* rodent experiments. It serves as both a preprocessing engine and a library to load and work with

data stored in capsules, embedding a copy of itself in each capsule. [software_name] generates the capsules with a single command line call, or two clicks on a browser-based interface (Figure 2a; Supplementary Figure S1a). One instance of the [software_name] browser interface can be run on a local server for an entire lab, processing many different experiment types and modalities with a queueing system, including Neuropixels recordings, two-photon imaging, behaviour, and optogenetics. It is built in a modular fashion which can be extended to other experiment types and new lab equipment using extensions that AI agents can write (Supplementary Figure S2; see Methods).

[software_name] facilitates collecting structured metadata. It includes customisable note-taking software to use while running experiments (Figure 2b), making these metadata and experimental observations standardised and automatically included in capsules. Capsules are discoverable through the browser interface for viewing (Figure 2c) and searching (Supplementary Figure S1b) experiment metadata, also accessible to agents via Model Context Protocol (MCP).

Capsules generated by [software_name] are versatile. Every capsule is structured as [redacted]. [software_name] loads capsules directly without the need to [redacted]. It also uses the flexibility of the capsule format to provide other conveniences, including lazy loading for quick startup times, and several specialised forms of compression to ensure small file sizes. [software_name] data capsules are typically about 50 MB for two-photon imaging experiments and 20 MB for Neuropixels recordings.

[software_name] can be quickly adapted to new laboratories, equipment, and experiment types. Adding a new piece of equipment or experiment type can be accomplished by writing an extension and placing it in a shared directory. Extensions can be written autonomously with agentic AI using bundled SKILL.md files. [software_name] also allows creating [redacted] that can recover data from anomalies in data acquisition, such as a software crash in the middle of an experiment. [software_name] itself was originally written largely without the use of AI; all routines were checked carefully for correctness both by hand and using AI assistance. [software_name] has accumulated over 14 person-years of use across seven researchers.

[software_name] data capsules enable agentic AI science

[software_name] can be used to locate and analyse experiments. We prompted the agent to find an experiment from a vague description and then generate a figure that required integrating data from multiple sources (Figure 3). Specifically, our prompt required the agent to find a specific experiment by searching metadata through the [software_name] MCP server. It then had to [redacted] The agent was able to locate the correct experiment and write code to produce the requested plot; the plot is shown here with only minor cosmetic adjustments (Figure 3). We closely inspected the code manually and with agent assistance, and found no conceptual or methodological errors. This demonstrates that agents can automatically locate capsules and use them to perform complex, multifaceted analyses.

[software_name] also enables agents to recover data from a failed experiment. We prompted the agent to recover data from a [redacted] experiment in which one of the acquisition computers crashed during the experiment. The agent was able to successfully generate a capsule from these data (Supplementary Figure S3a). We compared neural responses in the resulting capsule to another successful [redacted] experiment from the same session of the same animal. We found that cells had largely similar [redacted], indicating that the recovery process was successful (Supple-

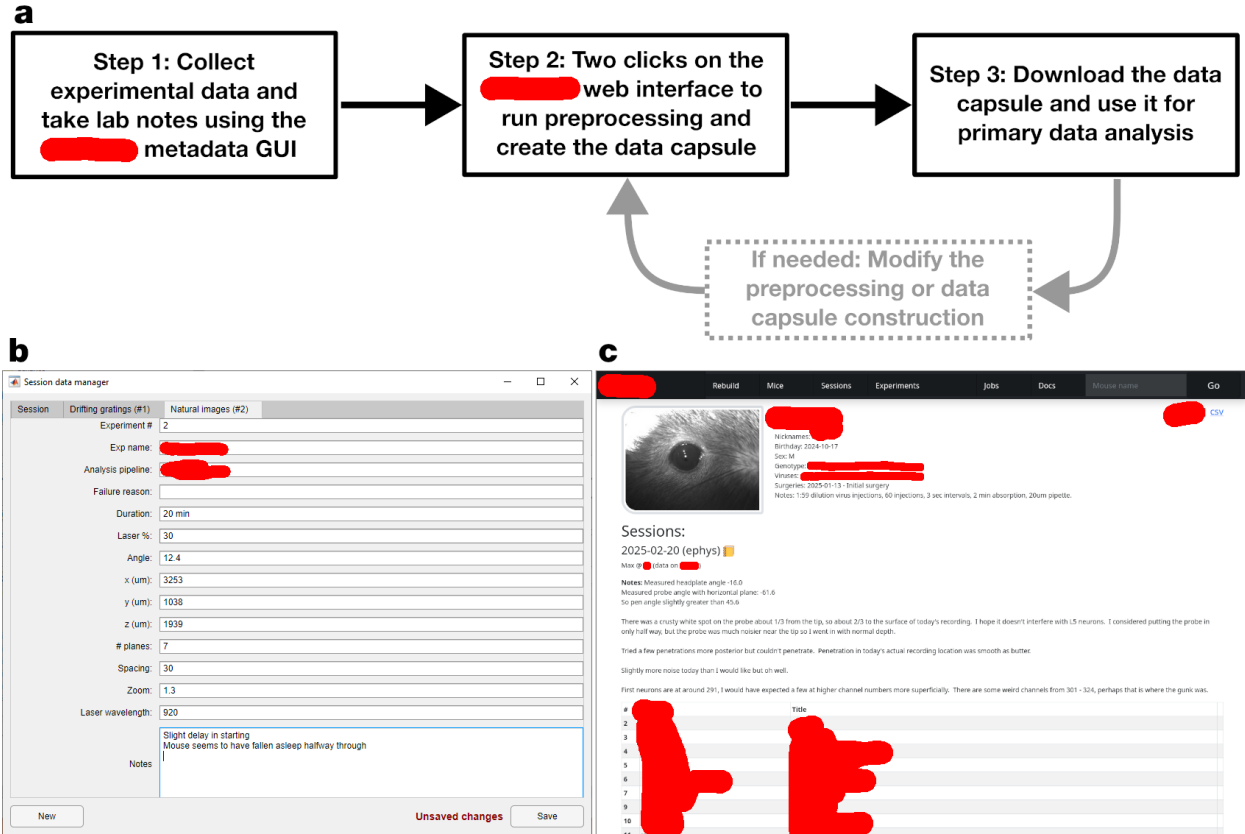


Figure 2: [software_name] workflow and interfaces. a. Schematic workflow for [software_name]. **b.** A graphical interface for collecting metadata during data collection. **c.** Browsing and searching experiment metadata on the browser interface.

mentary Figure S3b). Therefore, failed experiments can be recovered with [software_name] using the context provided in data capsules.

AI agents use a capsule's context to interpret data

Do agents perform better when they can access the context provided by a data capsule? We performed experiments to assess the importance of context in neuroscience data analysis. Systems neuroscience data typically include [redacted]. Data are varied and can include [redacted].

We collected [redacted]. Then, we used [software_name] to generate a data capsule for the experiment. We also created two redacted variants of the capsule by removing either contextual information alone, or contextual information and metadata (see Methods). Then, we assessed the ability of OpenAI GPT-6 to answer [redacted] questions about the data, 10 trials each, including [redacted] (Table 1). Questions were designed to assess the ability to perform analyses accurately and deduce information that would be helpful for interpretation. For each question, the AI agent [redacted]. Then, [redacted].

When data capsules were provided to the agent, it correctly answered almost all questions. It confidently and correctly answered all [redacted] questions each time (Figure 4a), making no mis-

Prompt:

```
>>> I recorded a [redacted]
[redacted] experiment a couple years
ago (late 2024 I think?), all I
remember is I [redacted]
[redacted]. For this session, show
all neurons from plane 3 on the field
of view image, coloured by their OSI
with brightness given by their
correlation with pupil size. Include
a scale bar. Format it as a panel for
a multipanel scientific figure.
```

Response:

```
Found [redacted]
[redacted].

The panel shows all [redacted] neurons in
Suite2p plane 3 (zero-based). Colour
represents vector OSI; brighter cells
have more positive pupil correlation.
Scale bar: 100  $\mu$ m.
```

Output:

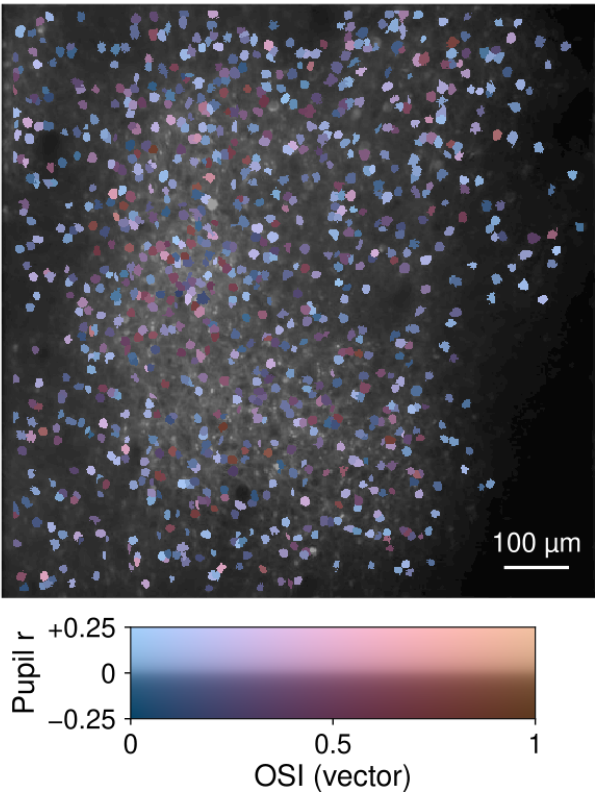


Figure 3: Agentic AI analysis with a [software_name] data capsule. The agent was prompted to locate an experimental session and produce a figure that integrates multiple types of data. The verbatim prompt and the agent’s truncated response are shown on the left, and the generated plot is shown on the right. [redacted]

Table 1: Questions and reference answers used to evaluate the importance of context. ROI, region of interest.

[redacted]

takes (Figure 4b), and correctly labelled the [redacted] (Figure 4c). The agent was less confident about its answers for questions related to [redacted], but was able to guess the correct answer to almost all questions (Figure 4d,e). There was only one exception: [redacted]

When context was redacted from the capsule, our agent had much more trouble interpreting the experiment. While it was able to correctly and confidently [redacted], it struggled with [redacted]. Notably, it failed to [redacted] (Table S1). It also [redacted]. [redacted]

A less powerful AI agent showed similar patterns but with more errors, more guesses, and fewer questions deemed unanswerable. We repeated these experiments with OpenAI GPT-5.6. With full context, as before, the agent was able to correctly and confidently answer many questions, correctly [redacted], and correctly [redacted] (Supplementary Figure S4). Without context, the

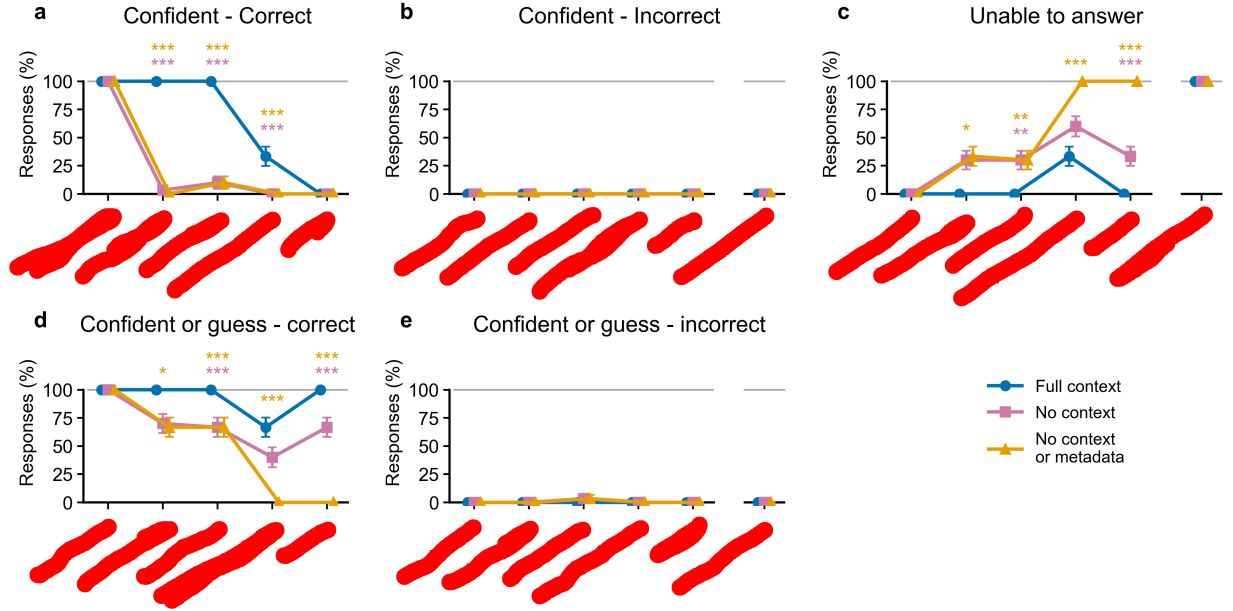


Figure 4: Context improves the accuracy of AI agents. AI agents attempted to answer the questions from Table 1, with 10 attempts at each question, using either a full data capsule or a data capsule with redacted context. **a,b.** Correct and incorrect responses when agents believed they had enough evidence to answer confidently. **c.** Responses deemed unanswerable. **d,e.** Correct and incorrect responses including both confident answers and guesses. The fraction of responses for each category is shown, using all [redacted] responses across [redacted] questions as the denominator. Colours indicate different context conditions. Error bars indicate Bernoulli SEM. [redacted] are omitted from the correct-answer panels and shown separately in the remaining panels. Errors were rare for GPT-6; full accuracy for each question is given in Supplementary Table S1. Stars indicate differences between full context and either no context (pink) or no context or metadata (yellow), using two-sided exact permutation tests with Bonferroni–Holm FWER correction across all 56 comparisons. Adjusted $p < 0.05$, $p < 0.01$, and $p < 0.001$ are indicated by one, two, and three stars, respectively.

agent [redacted]. The less powerful agent was overall more likely to [redacted]. Therefore, access to the full data capsule made all agents more accurate and confident.

Analysis of uncertain responses

For each question, we asked the AI to provide an explanation of how it came to its answer. We analysed the responses in order to find how the context provided by the capsule, or the lack of such context in redacted capsules, was driving its decisions.

[redacted]

Discussion

Data capsules make inspection of the source, preprocessing choices, and intended use of data easy. This is important to ensure that data can be interpreted correctly and confidently. We showed that adding experimental context improved the accuracy of the agent by increasing the accuracy of its guesses. While AI brings attention to the issue of correctly interpreting data due to the high-throughput analysis it enables, insufficient context can also be a problem for human collaborators; in situations where an AI might hallucinate or guess, humans often impose false assumptions of their own [4, 29] or just give up [30]. Another similarity between human and AI analysis is that both [redacted]. This can be assisted by automating capsule construction and using the same capsule for primary data analysis as for widespread release.

If constructed properly, any capsule found on the internet, a lab server, or even one's own hard drive, can be correctly analysed and understood without further information. Of course, this relies on a data capsule containing all of the necessary metadata in addition to the code. We showed that [software_name] capsules including [redacted] increased the ability of the AI agent to interpret the data. In human behavioural experiments, for example, it may be more useful to include the experimental protocol, template consent forms, computerised tasks, or preregistrations. These set the bounds on how deeply an AI agent can analyse the data and interpret the results.

Capsules provide both specificity and flexibility. Standards such as brain imaging data structure (BIDS) [2] and open neurophysiology environment (ONE) [31] provide general schemas for organising data files, whereas others such as neurodata without borders (NWB) [3, 32] and Connectivity Informatics Technology Initiative (CIFTI) [33] provide specific structures for specific types of data and metadata. While [software_name] uses its own custom format, other capsules could be based on these existing standards. For example, a capsule may consist of a zip file containing an NWB file together with the scripts used to generate it, and additional metadata about the experiment. Alternatively, an NWB file may operate as a capsule itself if it contains the necessary extra information, which could be included via an extension or its documented scratch space. These standard formats are designed to be programming language agnostic, whereas data capsules embrace language-specific algorithms and code, which can be trivially converted to other languages with AI agents.

The ideas behind data capsules come directly from best practices in software engineering. Keeping code and data together is a core tenet of object oriented programming, the software development style underlying the vast majority of commercial software [34]. Automatically generating capsules mirrors the concept of Makefiles, a widely used methodology to reduce human error and cognitive load [35]; Makefiles have recently become more common in science [27, 36, 37]. Additionally, using the same capsule for analysis and widespread distribution is similar to "dogfooding", a practice whereby software engineers heavily use the software they create, helping them find bugs and encouraging them to make it easier to use; dogfooding is famously used to improve the quality of software at Microsoft [38–40], Google [41], and other major companies.

Data capsules may not be appropriate for every situation. For example, large well-funded projects may have more resources to carefully curate their data and code by hand and provide a well-documented API to efficiently access them [31, 42–45]. However, this level of detail is out of reach for the typical researcher, and still prone to mistakes [28, 46, 47]. Data capsules may be more difficult to apply to sensitive human data, where federal laws and ethics boards limit which data may be accessed and by whom. In these cases, the original researchers may be authorised

to analyse more data than they are authorised to release, which could make capsules impractical.

When performing follow-up analyses on open data, the full responsibility of producing correct results—and the full credit for performing the analysis—is currently put on the shoulders of the analyst. By unwritten custom, only non-open data require a collaboration with the original experimenter. However, we showed that the accuracy and reasoning ability of agentic AI analysis depend greatly on how data are packaged to be shared. It is therefore highly advantageous for analysts working with open data to involve the original experimenter as a collaborator.

Methods

[redacted]

Data and code availability

The [software_name] source code, experimental data capsules, capsules containing the agent runs, and analysis scripts will be made public on journal acceptance.

AI Disclosure

OpenAI Codex was used to create a first draft of the methods section. It was also used to check for typos and awkward phrasing in the full manuscript. Code for [software_name] was largely written by hand, but OpenAI Codex was used to help debug and to assist with refactoring using a detailed human-created refactoring plan. Codex drafted much of the code used to perform the experiments in Figure 4 and Supplementary Figure S4. All text and code were carefully reviewed and checked by the author.

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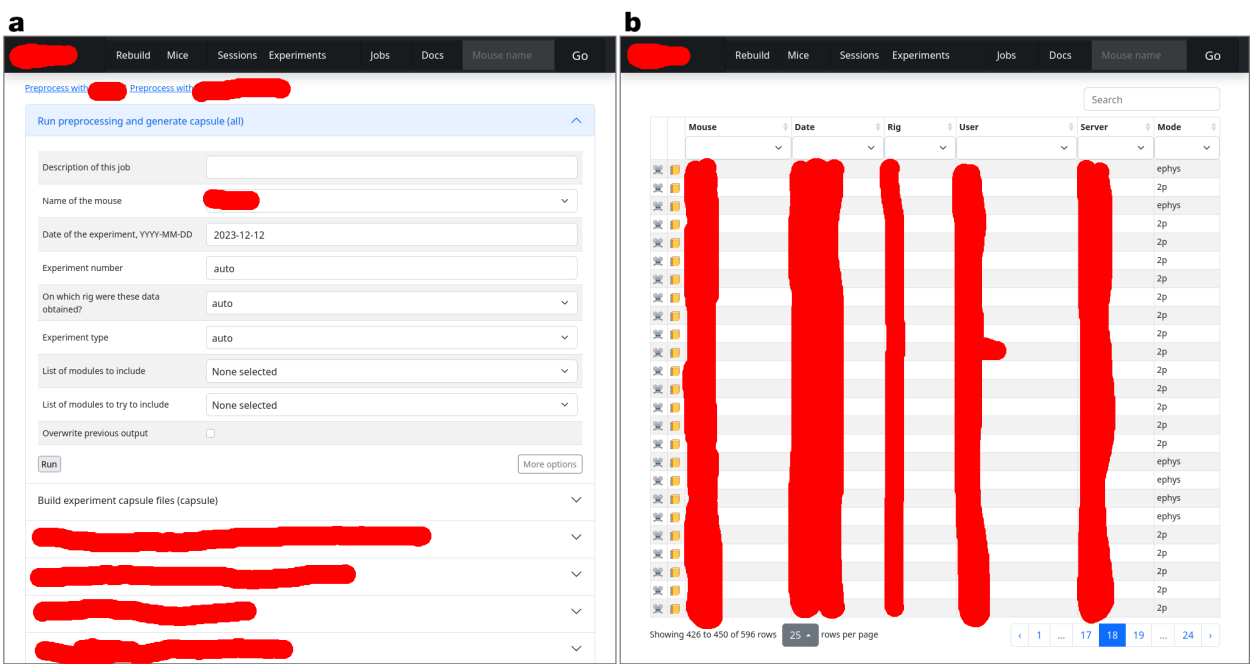
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Supplementary figures



Supplementary Figure S1: [software_name] browser interface. **a.** Generate capsules and run preprocessing pipelines. If multiple capsules are requested, they are added to a jobs queue. **b.** Browse and search experimental sessions. Similar browse and search functionality exists for finding mice and experiments.

[redacted]

Supplementary Figure S2: [software_name] implementation of a data capsule. [redacted]

a
Prompt:

```
>>> My [redacted]  
experiment terminated early. Write a  
patch to recover the data.
```

Response:

```
I need these inputs to write the recovery  
patch:
```

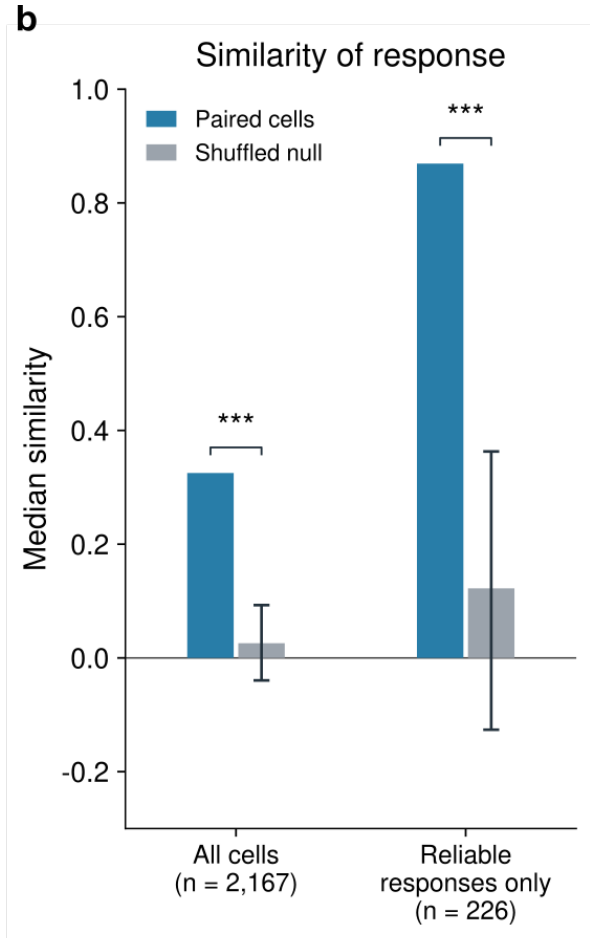
- Mouse, date, and experiment number.
- How it stopped, plus the complete [redacted] error traceback.
- Local copies of [redacted].mat and either [redacted].mat [redacted] or [redacted].mat [redacted].

Prompt:

```
>>> Okay, copied the files
```

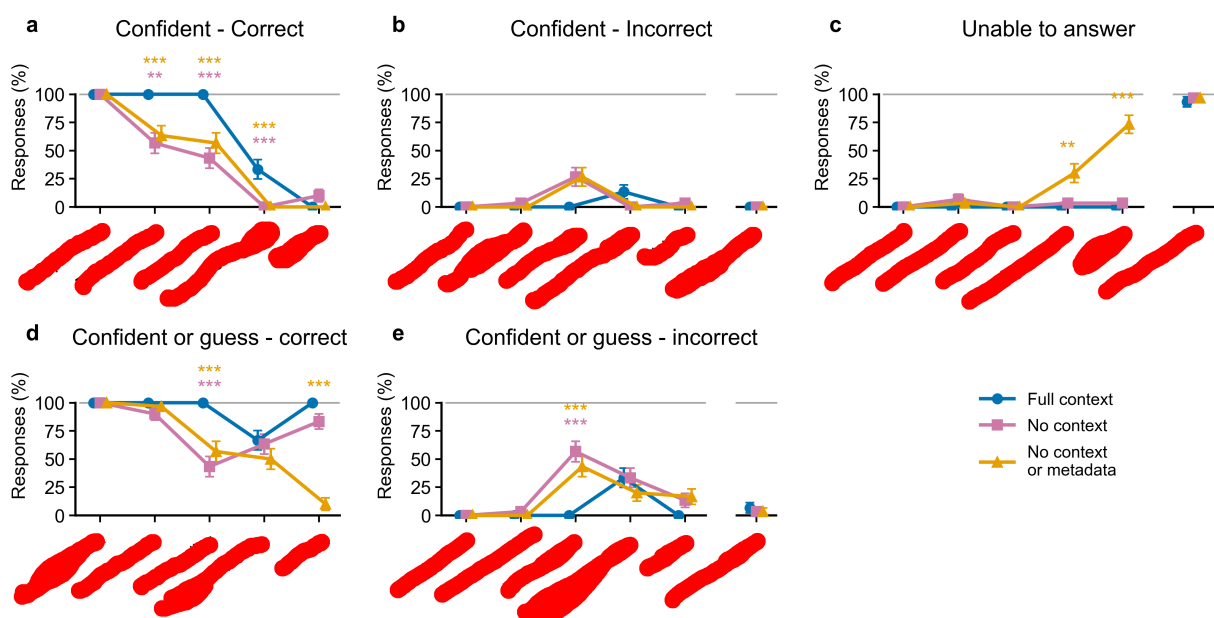
Response:

```
Created [redacted].py. It  
adds the missing interruption marker in  
memory, recovering 141 completed trials  
and excluding the unmatched final start.  
Raw files remain unchanged.
```



Supplementary Figure S3: Agentic AI recovery of a failed experiment with [software_name].

a. Verbatim prompts and truncated agent responses when asked to recover a [redacted] experiment that terminated early. The agent [redacted] to recover completed trials and exclude the incomplete trials. **b.** Median similarity of neural responses between the recovered experiment and a successful experiment from the same session. Responses from paired cells are compared with a shuffled null, using all cells ([redacted]) or only cells with reliable responses ([redacted]).



Supplementary Figure S4: Question accuracy for GPT-5.6. a-e. The same as Figure 4 except using GPT-5.6 instead of GPT-6. Full response accuracy is given in Supplementary Table S1.

Supplementary tables

Supplementary Table S1: **Correct and incorrect responses by question and context.** Counts underlying Figure 4 and Supplementary Figure S4. Each table cell gives correct/incorrect counts for [redacted] Full questions are given in Table 1.

[redacted]

Supplement 1: Suggested contents of a data capsule

Suggested elements to include in a capsule, where relevant:

[redacted]

Supplement 2: Examples of agents using and misusing context

The following are examples of the agent’s self-reported reasons for choosing an answer, selected to illustrate the behaviours discussed in the Results. Question numbers in the quotations may follow the randomised order presented to the agent, whereas question numbers in the source labels follow the canonical order in Table 1. Omitted text is indicated by [...] in the quotations.

Supplement 2.1: [redacted]

[redacted]

Supplement 2.2: [redacted]

[redacted]

Supplement 2.3: [redacted]

[redacted]

Supplement 2.4: [redacted]

[redacted]

Supplement 2.5: [redacted]

[redacted]

Supplement 2.6: [redacted]

[redacted]

Supplement 2.7: [redacted]

[redacted]

Supplement 2.8: [redacted]

[redacted]

Supplement 2.9: [redacted]

[redacted]